

I Have the Title of Data Analyst...Now What?!

A Practical Guide for New Fire &
Emergency Service Data
Professionals

Presented By:

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Introduction

I work for the City of Manassas (VA) Fire and Rescue Department. We are in Northern Virginia, not far from where we are today in Fairfax County.

The Department was formed in 2008 as a combination system consisting of one career department, two volunteer organizations: Manassas Volunteer Fire Company (1892) and the Greater Manassas Volunteer Rescue Squad (1966), and two stations (plus a separate Headquarters office).

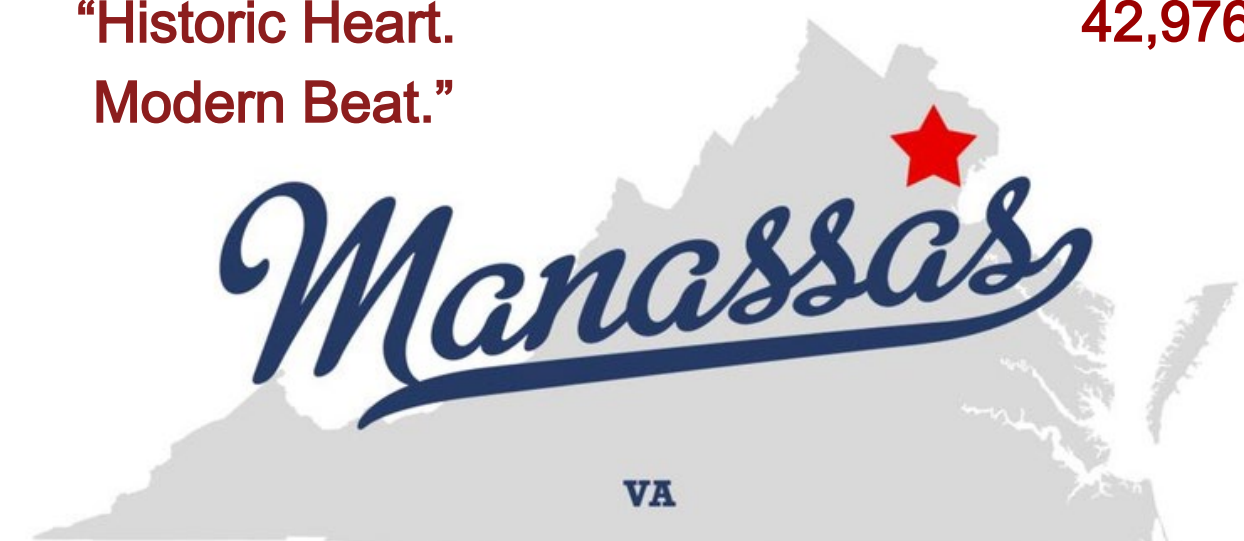
Today we have grown to 74 positions (only 3 of which are civilian).

A little background on Manassas VA

Manassas still has a large railroad presence with trains from Virginia Railway Express, Amtrak, and Norfolk Southern. We have a regional airport that is in the process of going commercial; one hospital, a few data centers (and growing), and tech giants like Micron, and BAE Systems.

Our City motto is:
"Historic Heart.
Modern Beat."

Population of
42,976



★ Designed by TownMapsUSA.com

That's all great, but how did I get the responsibilities of a Data Analyst?

That lifechanging question my Chief asked me was “What do you think about tackling the data for the department?”

A few problems though....

We did not have access to our data

Prince William County allows us to run on their CAD and dispatch, and therefore have our raw data. Once we began to look at our data, we had to make a plan with their staff as to how we could obtain our data often. They have been wonderful in helping us!!!

I didn't know how to be an analyst

I had no idea where to start, even after we received our data. My Chief agreed to have my department pay for a class.

I immediately began looking into classes, and later started with Google / Coursera online classes.

Imposter Syndrome is REAL!



Why Does Data Matter?



Improves Outcomes

Drive faster response times, enhance crew safety, and reduce property loss through data-informed decisions



Aligns to Standards

Track performance against NFPA 1710 benchmarks and define service level objectives clearly



Support Strategic Planning

Optimize staffing models, deployment plans, station locations, and apparatus replacement schedules



Ensures Accountability

Provide transparent reporting to City Council, community stakeholders, and for accreditation/grant purposes

Setting Expectations

One of the first things I learned was that I needed to have a long chat with my Chief and set expectations. I had so many questions, as did he. If we didn't talk about them early, the road could get very bumpy.

Define

What does data mean to your Chief?

What data/items does your Chief want you to track/analyze? Operations? EMS KPIs? Staffing?

Expectation(s)

Is your Chief expecting you to use specific software to analyze data and/or create dashboards?

Or, how do they want to see the data?

How often does your Chief want to see data from you? Daily, weekly, monthly?

Learning

Do you feel you need training/classes to best perform the role of Data Analyst?

If yes, is your Chief aware of this?

Training Inventory

Before looking for classes or training opportunities, take a quick inventory of what you already know.

Potential Skills

Excel

Basic how - to - use; Features, Formulas, Charts / Graphs, and Pivot Tables

SQL

Basic how to's - like what does SQL stand for? (Structured Query Language)

Tableau or PowerBI

Basic how to's for creating graphs and/or dashboards

Data Cleaning

How to handle missing data, inconsistencies, and organizing it properly

Communication(s) and Story Telling

What story are you trying to tell with your data, and who are you telling?

Potential Resources / Classes

Online (Ex. Coursera)

Google Certification classes | www.Coursera.org

Fire Stats

Data Analysis and Presentation for Fire and EMS | www.FireStats.com

College Certification Programs

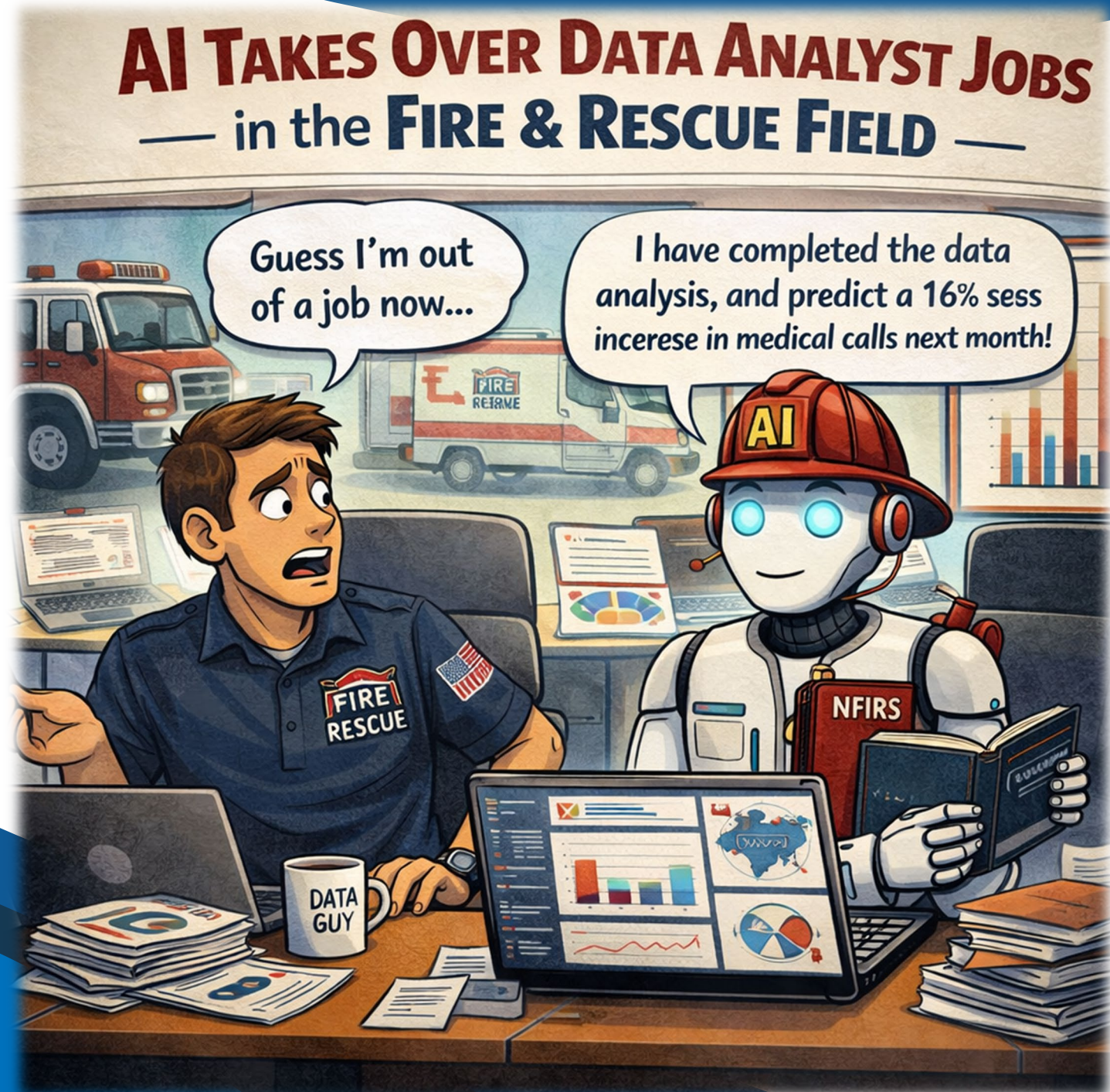
Many colleges have certification programs centered around Data Analytics

Conference(s)

RMS Vendor Conferences, Center for Public Safety Excellence, etc.

AFESA (Association of Fire & Emergency Services Analysts)

The AFESA group is full of tips, tricks, learning opportunities, and most importantly – helpful professionals



The Role of Artificial Intelligence:

It's the Great AI Takeover

For many years now, since I was a child, we've heard that AI was going to take over all of our jobs and there'd be no need for us. To make it worse, we hear

"AI" all the time in our world. What's happening?

RELAX...while AI is not taking over, it can be helpful in many ways.



Automate
Processes



Create Interactive
Dashboards

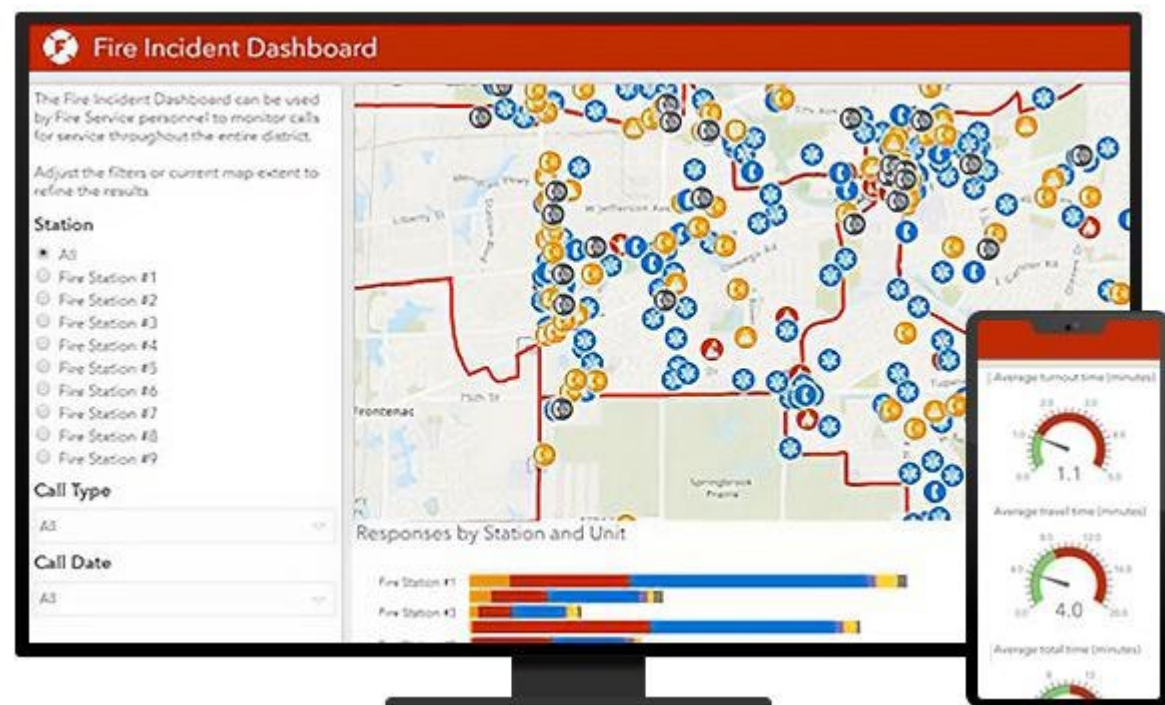


If you can
imagine it, AI
can probably
help make it
happen

Data for Beginners

While some Analysts can setup intricate dashboards and make sifting through the data look like the blink of an eye, it takes some of us a bit longer and that's OK! Do not get discouraged!

Data, at it's core, can be broken down into a few steps.



✓ Collect Data

✓ Data Dictionary

✓ Clean the Data

✓ Analyze the Data

✓ Report Clearly

Collect Data

The Fire and Rescue Service obtains data from many different sources:

- Fire Reports
- EMS Reports
- CAD
- KPIs (Key Performance Indicators)
- Budget Metrics

Where does your data come from?

Do you currently have access to the raw data, or do you need to get that?



A blue and white fire truck from the City of Manassas is shown. The side of the truck features the text 'CITY OF MANASSAS' and a circular seal. The truck is parked on a street. Overlaid on the right side of the image is a semi-transparent blue rectangle containing the text 'Collect Data' and a table of incident data.

Collect Data

Callsign	Incident Number	Report Number	Incident Type	Code	Location	Box	Latitude
A521	FD25123000077232	MNSS250006776	INJ	B	LEE AVE	50157	38.75
A521	FD25123000077243	MNSS250006778	SIC	A1	GEORGIAN CT	50119	38.77
A521	FD25123000077255	MNSS250006779	ACC	B	LIBERIA AVE	51650	38.74
A521	FD25123000077260	MNSS250006781	SIC	A1	LIBERIA AVE	51650	38.74
A521	FD25123000077265	MNSS250006782	SIC	B	JACKSON AVE	50116	38.76
A521	FD25123100077275	MNSS250006787	INJ	B	DUMFRIES RD	50723	38.72

Inspection history for Vehicle & Station checks

Show/Hide Search Settings

☒ Vehicle/Station ☐ Department Checklist

ALS Bag 1 ▼

All ▼

All

All ▼

All ▼

Wed Apr 1, 2026 to Wed Apr 15, 2026

[VIEW SEARCH RESULTS](#)

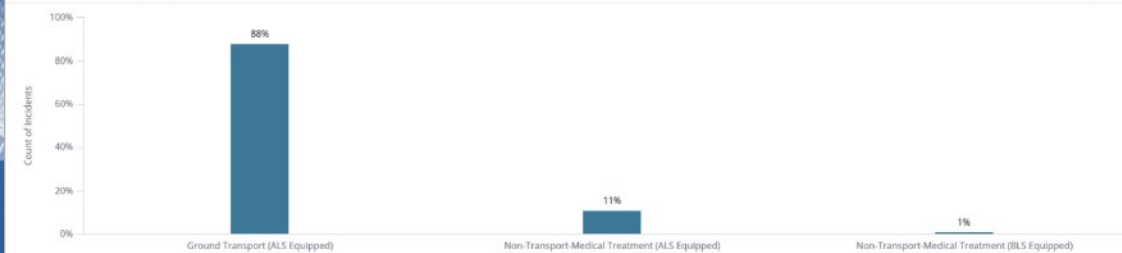
Results will display below, with PDF & Excel export options

ALS vs BLS Unit Response Breakdown

This report looks at the breakdown of responses by Unit Capability (eResponse.07). Please note this field was introduced with the transition to NEMESIS 3.5. Unit Level of Care (eResponse.15) was retired with NEMESIS 3.4.

Analyze It ⓘ 🔗

Total incidents: 379

Time in Dispatch	Incident Number	Disposition/NEMIS 3.4i	Unit Disposition	Patient Evaluation/Care Disposition	Crew Disposition
02/23/2026 11:27:45	MN5526009053	Cancelled (Prior to Arrival at Scene)	Cancelled Prior to Arrival at Scene	NA	Back in Service. No Care or Support Ser
02/23/2026 11:01:55	MN5526009052	Transported Lights/Siren	Patient Contact Made	Patient Evaluated and Care Provided	Initiated and Continued Primary Care
02/23/2026 09:50:27	MN5526009050	Transported No Lights/Siren	Patient Contact Made	Patient Evaluated and Care Provided	Initiated and Continued Primary Care
02/23/2026 09:37:32	MN5526009047	Transported No Lights/Siren	Patient Contact Made	Patient Evaluated and Care Provided	Initiated and Continued Primary Care
02/23/2026 22:36:43	MN5526009046	Transported Lights/Siren	Patient Contact Made	Patient Evaluated and Care Provided	Initiated and Continued Primary Care
02/23/2026 21:49:28	MN5526009045	Patient Refused Evaluation/Care (Without Transport)	Patient Contact Made	Patient Evaluated and Care Provided	Back in Service. Care or Support Ser
02/23/2026 19:09:59	MN5526009043	Transported No Lights/Siren	Patient Contact Made	Patient Evaluated and Care Provided	Initiated and Continued Primary Care

Callsign	Incident Number	Report Number	Incident Type	Code	Location	Box	Latitude	Longitude	Dispatch Time	Enroute Time	On Scene Time	Enroute Hospital Time	Arrive Hospital Time	Clear Time
A521	FD25123000077232	MNSS250006776	INJ	B	LEE AVE	50157	38.75206425	-77.4800068	12/30/2025 7:22:43 PM	12/30/2025 7:23:51 PM	12/30/2025 7:31:12 PM	12/30/2025 7:55:43 PM	12/30/2025 8:03:07 PM	12/30/2025 8:24:52 PM
A521	FD25123000077243	MNSS250006778	SIC	A1	GEORGIAN CT	50119	38.77168805	-77.4846783	12/30/2025 9:13:10 PM	12/30/2025 9:14:34 PM	12/30/2025 9:22:52 PM			12/30/2025 9:48:31 PM
A521	FD25123000077255	MNSS250006779	ACC	B	LIBERIA AVE	51650	38.74253565	-77.4495304	12/30/2025 10:08:20 PM	12/30/2025 10:08:38 PM	12/30/2025 10:14:26 PM			12/30/2025 10:34:24 PM
A521	FD25123000077260	MNSS250006781	SIC	A1	LIBERIA AVE	51650	38.74440215	-77.4488261	12/30/2025 10:37:04 PM	12/30/2025 10:37:12 PM	12/30/2025 10:40:02 PM			12/30/2025 11:25:47 PM
A521	FD25123000077265	MNSS250006782	SIC	B	JACKSON AVE	50116	38.76076618	-77.4733753	12/30/2025 11:44:12 PM	12/30/2025 11:45:35 PM	12/30/2025 11:52:24 PM	12/31/2025 12:27:33 AM	12/31/2025 12:44:53 AM	12/31/2025 12:49:52 AM
A521	FD25123100077275	MNSS250006787	INJ	B	DUMFRIES RD	50723	38.72619681	-77.4718133	12/31/2025 1:49:28 AM	12/31/2025 1:51:21 AM	12/31/2025 1:59:42 AM			12/31/2025 2:17:09 AM

Data Dictionary

What is it?

By definition, at least according to Google, “a data dictionary is a centralized, foundational repository of metadata that defines the structure, meaning, characteristics, and relationships of data elements within a database or system.”

Data Dictionaries allow us to define, document, and manage data elements for consistent, accurate use.

Column Name	Meaning
Callsign (grouping)	Grouped by Unit
Callsign*	Unit Number/ID (501, 521, and 581 are City of Manassas Units)
Incident Number	Incident Number (unique to each incident, but not unique to each unit that responds)
Report Number	Incident Number (unique to each incident, but not unique to each unit that responds)
Incident Type	Type of Incident (ex. ACC = accident; INJ = injury)
Code	Goes along with the type of incident (ex. A = ALS; B = BLS)
Location	Address of Incident
Box*	Box Number
Latitude	Latitude of Incident
Longitude	Longitude of Incident
Dispatch Time	Time the call was dispatched by PWC PSCC
Enroute Time	Time units marked as responding/enroute
	<i>The difference between the Enroute Time and Dispatch Time is the “CHUTE TIME” (how long it took units to hear the call, get into the unit, and proceed to respond)</i>
On-Scene Time	Time units marked on-scene, or arrival at incident location
	<i>Difference between On-Scene Time and Dispatch Time is the “RESPONSE TIME”</i>
Enroute Hospital Time	Time units left the incident location and began transporting to the hospital
Arrive Hospital Time	Time units arrived at the hospital with a patient
Clear Time	Time units cleared the incident
	<i>IF no patients are transported, Clear Time minus On-Scene Time will be the “TOTAL TIME ON-SCENE.” Otherwise, you could use Clear Time minus Dispatch Time for “TOTAL CALL TIME.”</i>
Enroute Hospital Time to Arrive Hospital Time in Seconds	Amount of time it took for units to transport from the incident location to the hospital, in seconds
Arrive Hospital Time to Cleared Time in Seconds	Amount of time it took units to clear from the hospital after arriving with a patient, in seconds
Enroute Time to On Scene Time in Seconds	Total Response Time (On-Scene Time minus Enroute Time), in seconds
Dispatch Time to Cleared Time in Seconds	Total Call Time (Cleared Time minus Dispatch Time), in seconds

Example of my own data dictionary

Incident - Aid Type [🔗](#)

The type of aid given or received.

modules: [Core Incident](#) type: text element name: incident_aid_type

Values [↕](#)

Incident - Aid Direction [🔗](#)

Whether aid was given or received.

modules: [Core Incident](#) type: text element name: incident_aid_direction

Values [↕](#)

Incident - Actions and Tactics [🔗](#)

Actions taken by the fire department on the scene of the incident.

modules: [Core Incident](#) type: text element name: incident_actions_taken

Values [↕](#)

And... the data dictionary from NERIS

Data Dictionary

50100	COM
50101	COM
50102	COM
50103	COM
50104	COM
50105	COM
50106	COM
50107	COM
50108	COM
50109	COM
50110	COM
50111	COM
50112	COM
50113	COM
50114	COM
50115	COM
50116	COM
50117	COM
50118	COM

50414	PWC
50415	PWC
50416	PWC
50417	PWC
50418	PWC
50419	PWC
50420	PWC
50421	PWC
50422	PWC
50423	PWC
50424	PWC
050425K	PWC
050426K	PWC
50427	PWC
50428	PWC



Cleaning data allows for...

Format data precisely for enhanced accuracy and improved usability



Duplicates

Do you see anything in the photo below that could be cleaned up for more accurate count of total call volume?

Anything else that should be “cleaned?”

Callsign	Incident Number	Report Number	Incident Type	Code
5C	FD26011400002776	FD260001935	INGAS	COMM
5C	FD26011800003641	FD260002567	INGAS	HOUSE
5D	FD26011500002951	FD260002052	STRUCT	COMM
5E	FD26011500002951	FD260002052	STRUCT	COMM
5F	FD26011500002951	FD260002052	STRUCT	COMM
5G	FD26011500002951		STRUCT	COMM

Data Cleaning

Data Cleaning is defined, by Google, as “the process of identifying and fixing or removing incorrect, incomplete, duplicate, or improperly formatted data within a dataset to improve its quality.”



Data Cleaning

Incident Type	Dispatch Time	Enroute Time	CHUTE TIME	CHUTE TIME (TEST)	On Scene Time	RESPONSE TIME	RESPONSE TIME (TEST)	Enroute Hospital Time	Arrive Hospital Time	Clear Time
FALARM	8/18/2025 7:51:54 AM	8/18/2025 7:53:35 AM	0:01:41	0:01:41		#####	N/A			8/18/2025 7:58:52 AM
FSERV	8/18/2025 9:59:18 AM	8/18/2025 10:00:12 AM	0:00:53	0:00:53	8/18/2025 10:07:52 AM	0:08:34	0:08:34			8/18/2025 10:10:28 AM
RESCUE	8/18/2025 10:17:49 AM	8/18/2025 10:19:13 AM	0:01:24	0:01:24		#####	N/A			8/18/2025 10:20:02 AM
RESCUE	8/18/2025 10:19:53 AM	8/18/2025 10:20:06 AM	0:00:13	0:00:13	8/18/2025 10:23:20 AM	0:03:27	0:03:27			8/18/2025 10:39:36 AM
SIC	8/18/2025 5:52:07 PM	8/18/2025 5:53:20 PM	0:01:12	0:01:12	8/18/2025 5:56:37 PM	0:04:30	0:04:30	8/18/2025 6:27:13 PM	8/18/2025 6:36:22 PM	8/18/2025 7:15:20 PM
SIC	8/18/2025 5:52:07 PM	8/18/2025 5:53:35 PM	0:01:27	0:01:27	8/18/2025 5:56:50 PM	0:04:42	0:04:42			8/18/2025 6:22:07 PM
INJ	8/18/2025 6:03:54 PM	8/18/2025 6:04:47 PM	0:00:53	0:00:53	8/18/2025 6:07:18 PM	0:03:24	0:03:24	8/18/2025 6:17:41 PM	8/18/2025 6:20:10 PM	8/18/2025 6:42:37 PM
INJ	8/18/2025 6:03:54 PM	8/18/2025 6:04:53 PM	0:00:59	0:00:59	8/18/2025 6:07:33 PM	0:03:39	0:03:39			8/18/2025 6:16:34 PM
INGAS	8/18/2025 6:55:55 PM	8/18/2025 6:56:37 PM	0:00:42	0:00:42	8/18/2025 7:00:06 PM	0:04:11	0:04:11			8/18/2025 7:16:39 PM
INGAS	8/18/2025 6:55:55 PM	8/18/2025 6:57:24 PM	0:01:29	0:01:29	8/18/2025 6:59:49 PM	0:03:54	0:03:54			8/18/2025 7:27:22 PM
INGAS	8/18/2025 6:55:55 PM	8/18/2025 6:57:00 PM	0:01:05	0:01:05	8/18/2025 7:02:02 PM	0:06:07	0:06:07			8/18/2025 7:04:40 PM
INGAS	8/18/2025 6:55:55 PM	8/18/2025 6:56:46 PM	0:00:51	0:00:51	8/18/2025 6:59:52 PM	0:03:57	0:03:57			8/18/2025 7:26:17 PM
SIC	8/18/2025 7:12:34 PM	8/18/2025 7:14:11 PM	0:01:37	0:01:37	8/18/2025 7:21:54 PM	0:09:19	0:09:19	8/18/2025 7:38:20 PM	8/18/2025 7:49:48 PM	8/18/2025 8:09:30 PM
INJ	8/18/2025 7:54:16 PM	8/18/2025 7:54:50 PM	0:00:34	0:00:34	8/18/2025 7:59:29 PM	0:05:14	0:05:14		8/18/2025 8:26:01 PM	8/18/2025 8:50:09 PM
ASS	8/18/2025 9:11:29 PM	8/18/2025 9:12:49 PM	0:01:20	0:01:20	8/18/2025 9:21:53 PM	0:10:24	0:10:24	8/18/2025 9:41:37 PM	8/18/2025 9:47:28 PM	8/18/2025 10:12:11 PM
INJ	8/18/2025 9:20:29 PM	8/18/2025 9:22:13 PM	0:01:45	0:01:45	8/18/2025 9:26:50 PM	0:06:22	0:06:22	8/18/2025 9:37:45 PM	8/18/2025 9:44:59 PM	8/18/2025 9:57:11 PM
INJ	8/19/2025 12:26:15 AM	8/19/2025 12:28:30 AM	0:02:15	0:02:15	8/19/2025 12:29:41 AM	0:03:26	0:03:26			8/19/2025 12:44:49 AM
INJ	8/19/2025 12:26:15 AM	8/19/2025 12:28:09 AM	0:01:54	0:01:54	8/19/2025 12:29:25 AM	0:03:10	0:03:10			8/19/2025 12:38:14 AM
STOP	8/19/2025 2:26:32 AM	8/19/2025 2:29:02 AM	0:02:30	0:02:30	8/19/2025 2:34:40 AM	0:08:08	0:08:08	8/19/2025 3:06:58 AM	8/19/2025 3:13:09 AM	8/19/2025 3:42:51 AM
STOP	8/19/2025 2:26:32 AM	8/19/2025 2:28:39 AM	0:02:07	0:02:07	8/19/2025 2:32:51 AM	0:06:19	0:06:19	8/19/2025 3:06:49 AM	8/19/2025 3:13:16 AM	8/19/2025 4:25:52 AM
STOP	8/19/2025 2:26:32 AM	8/19/2025 2:29:28 AM	0:02:56	0:02:56	8/19/2025 2:32:12 AM	0:05:40	0:05:40			8/19/2025 3:35:36 AM
STOP	8/19/2025 2:36:38 AM	8/19/2025 2:37:20 AM	0:00:43	0:00:43	8/19/2025 2:40:13 AM	0:03:35	0:03:35			8/19/2025 3:07:41 AM
SIC	8/19/2025 2:44:56 AM	8/19/2025 2:47:42 AM	0:02:45	0:02:45	8/19/2025 2:53:17 AM	0:08:21	0:08:21			8/19/2025 3:01:43 AM
SIC	8/19/2025 7:07:42 AM	8/19/2025 7:08:53 AM	0:01:11	0:01:11	8/19/2025 7:11:13 AM	0:03:31	0:03:31		8/19/2025 7:40:52 AM	8/19/2025 8:32:50 AM
SIC	8/19/2025 7:07:42 AM	8/19/2025 7:08:38 AM	0:00:56	0:00:56	8/19/2025 7:12:57 AM	0:05:15	0:05:15			8/19/2025 7:29:00 AM
SIC	8/19/2025 9:01:47 AM	8/19/2025 9:03:01 AM	0:01:14	0:01:14	8/19/2025 9:08:08 AM	0:06:21	0:06:21			8/19/2025 9:27:51 AM
SIC	8/19/2025 9:02:20 AM	8/19/2025 9:02:37 AM	0:00:17	0:00:17	8/19/2025 9:07:09 AM	0:04:49	0:04:49	8/19/2025 9:23:23 AM	8/19/2025 9:28:18 AM	8/19/2025 10:00:08 AM
SIC	8/19/2025 9:02:20 AM	8/19/2025 9:03:59 AM	0:01:40	0:01:40	8/19/2025 9:10:26 AM	0:08:07	0:08:07			8/19/2025 9:18:40 AM

Look -up Fields

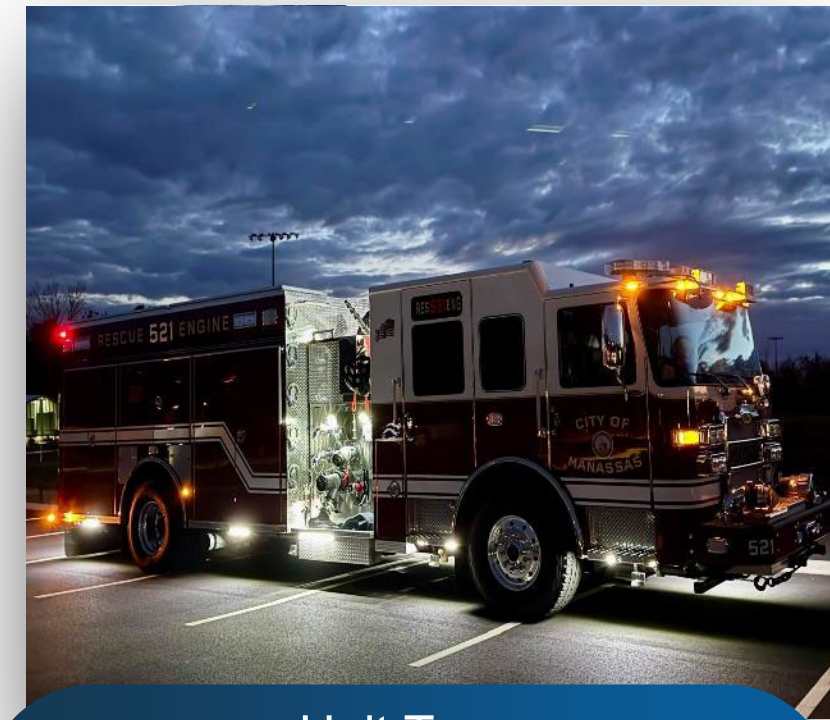
After the cleaning stage, and before analysis, I add my look -up fields. These fields help search through large datasets for key identifiers. A look -up formula compares a dataset to a range of data in the look -up fields and returns the appropriate information from other rows or columns.



Call Type(s)
EMS, Fire, Hazmat, etc.



What area or jurisdiction a
BOX NUMBER represents



Unit Type
Ambulance, Engine, Tower,
Bike, etc.

Vlookup Fields

Callsign	Incident Number
TW501	FD25081800048144
PE521	FD25081800048175
T511	FD25081800048176
TW501	FD25081800048176
M521	FD25081800048266
PE521	FD25081800048266
M501	FD25081800048271
PE501	FD25081800048271
BC581	FD25081800048282
PE501	FD25081800048282
PE521	FD25081800048282
TW501	FD25081800048282
M501	FD25081800048286
M521	FD25081800048292
A508	FD25081800048304
M501	FD25081800048305
M521	FD25081900048317
PE521	FD25081900048317
EMS581	FD25081900048324

FORMULA =VLOOKUP(A2,Sheet1!A\$1:C\$112,2,FALSE)

An example of my lookup field list is:

E501	Engine	COM
------	--------	-----

The formula will look at the “Callsign” then go to my lookup field list, find that callsign, and will then know what type of apparatus it is and what jurisdiction it belongs to.

Callsign	UNIT TYPE	JURISDICTION (Responding From)
TW501	Tower	COM
PE521	Paramedic Engine	COM
T511	Truck	PWC
TW501	Tower	COM
M521	Medic	COM
PE521	Paramedic Engine	COM
M501	Medic	COM
PE501	Paramedic Engine	COM
BC581	Battalion Chief	COM
PE501	Paramedic Engine	COM
PE521	Paramedic Engine	COM
TW501	Tower	COM
M501	Medic	COM
M521	Medic	COM
A508	Ambulance	PWC
M501	Medic	COM
M521	Medic	COM
PE521	Paramedic Engine	COM
EMS581	EMS Supervisor	COM

Vlookup Fields

Callsign	Incident Number
TW501	FD25081800048144
PE521	FD25081800048175
T511	FD25081800048176
TW501	FD25081800048176
M521	FD25081800048266
PE521	FD25081800048266
M501	FD25081800048271
PE501	FD25081800048271
BC581	FD25081800048282
PE501	FD25081800048282
PE521	FD25081800048282
TW501	FD25081800048282
M501	FD25081800048286
M521	FD25081800048292
A508	FD25081800048304
M501	FD25081800048305
M521	FD25081900048317
PE521	FD25081900048317
EMS581	FD25081900048324

FORMULA =VLOOKUP(A2,Sheet1!A\$1:C\$112,2,FALSE)

A2 = the original cell I want to look for
Sheet 1 = where my lookup information is (and then the cells where that data lives)
2 = the column within my lookup fields that I want to show
FALSE = I want an exact match
TRUE would give an approximate match

=VLOOKUP(lookup_value, table_array, col_index_num, [range_lookup])

An example of my lookup field list is:

E501	Engine	COM
------	--------	-----

The formula will look at the “Callsign” then go to my lookup field list, find that callsign, and match it within the lookup field list. Then it will know what type of apparatus the callsign is and what jurisdiction it belongs to.

Callsign	UNIT TYPE	JURISDICTION (Responding From)
TW501	Tower	COM
PE521	Paramedic Engine	COM
T511	Truck	PWC
TW501	Tower	COM
M521	Medic	COM
PE521	Paramedic Engine	COM
M501	Medic	COM
PE501	Paramedic Engine	COM
BC581	Battalion Chief	COM
PE501	Paramedic Engine	COM
PE521	Paramedic Engine	COM
TW501	Tower	COM
M501	Medic	COM
M521	Medic	COM
A508	Ambulance	PWC
M501	Medic	COM
M521	Medic	COM
PE521	Paramedic Engine	COM
EMS581	EMS Supervisor	COM

Look -up Fields

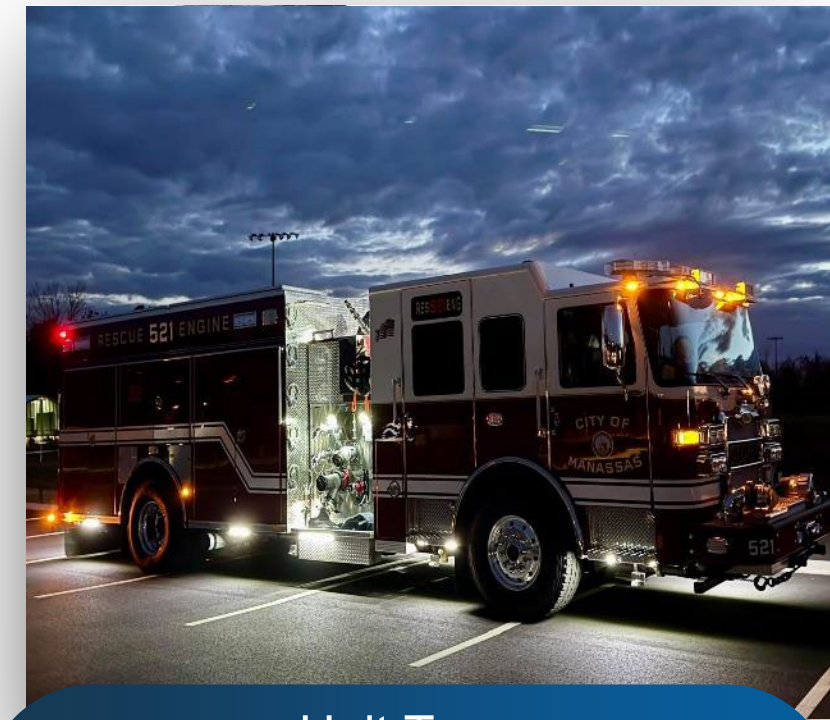
After the cleaning stage, and before analysis, I add my look -up fields. These fields help search through large datasets for key identifiers. A look -up formula compares a dataset to a range of data in the look -up fields and returns the appropriate information from other rows or columns.



Call Type(s)
EMS, Fire, Hazmat, etc.



What area or jurisdiction a
BOX NUMBER represents



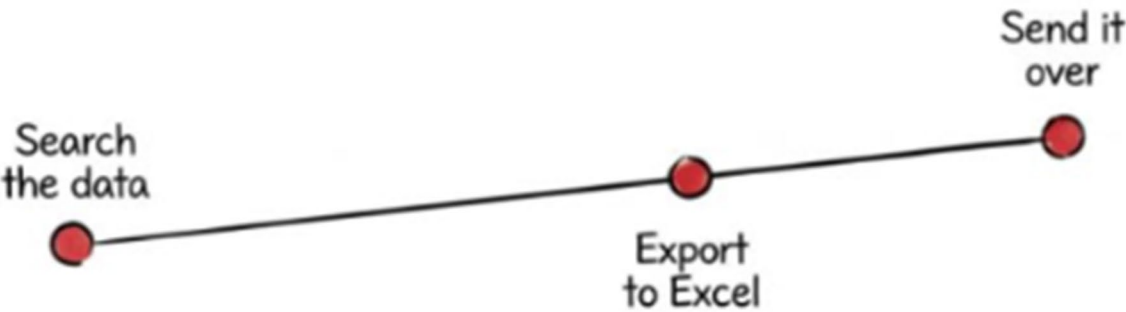
Unit Type
Ambulance, Engine, Tower,
Bike, etc.

Analyze the Data

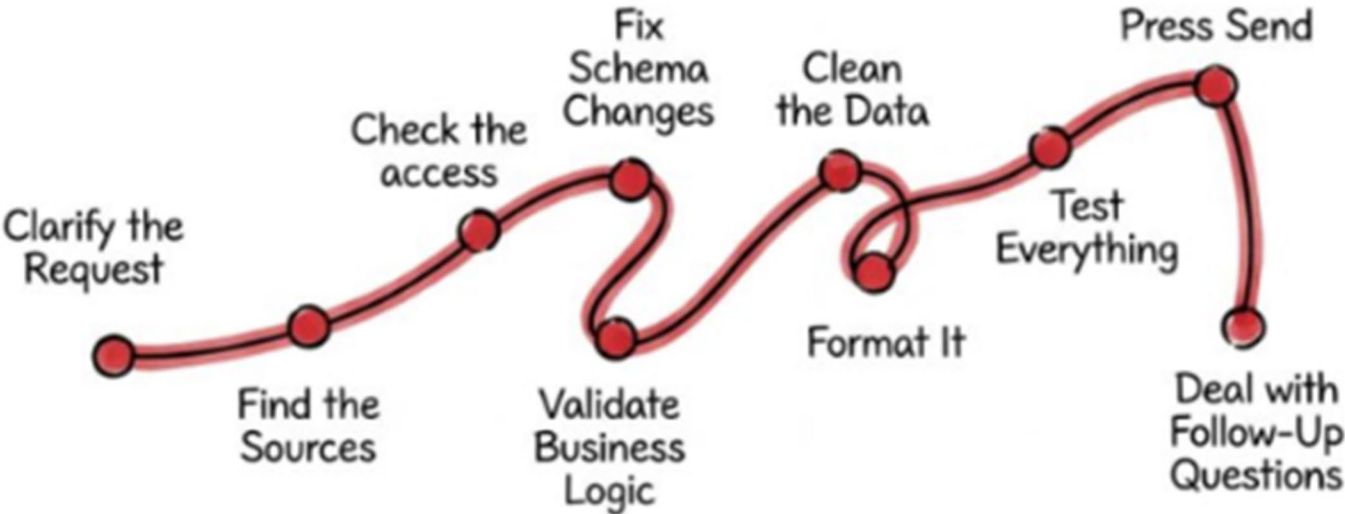
Data is analyzed to identify trends, patterns, and improvements. Some information will present immediately, while others will take time – a repeated pattern or trend.

This can be done via dashboards, or Excel spreadsheets.

What People **THINK** a **Quick Data Pull** Looks Like



What It **Really** **Looks Like**



Marcel Dybalski

Response Time

Response Time is currently a hot topic in my department so this is always a data point that we look at closely.

Total Call Volume, and Call Volume by Call Type

Of course we want, and need, to know what types of calls we are responding to, where, and when.

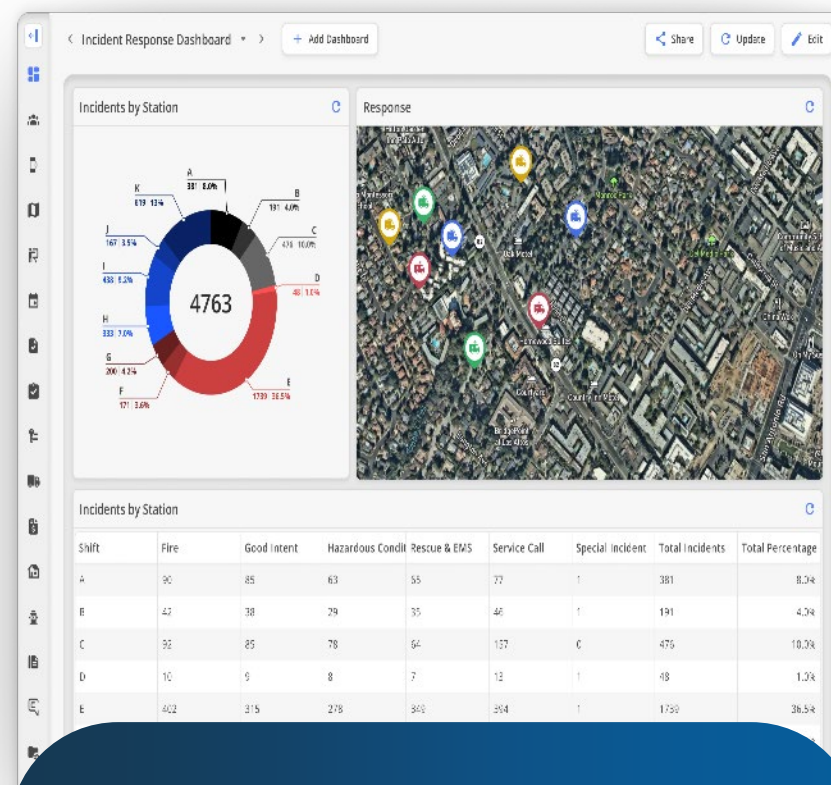
Dispatch Algorithms

In my jurisdiction we have been looking at the dispatch algorithms surrounding EMS calls. Should an ALS or BLS unit be dispatched; if BLS, where is the closest unit and are they being taxed physically and mentally?

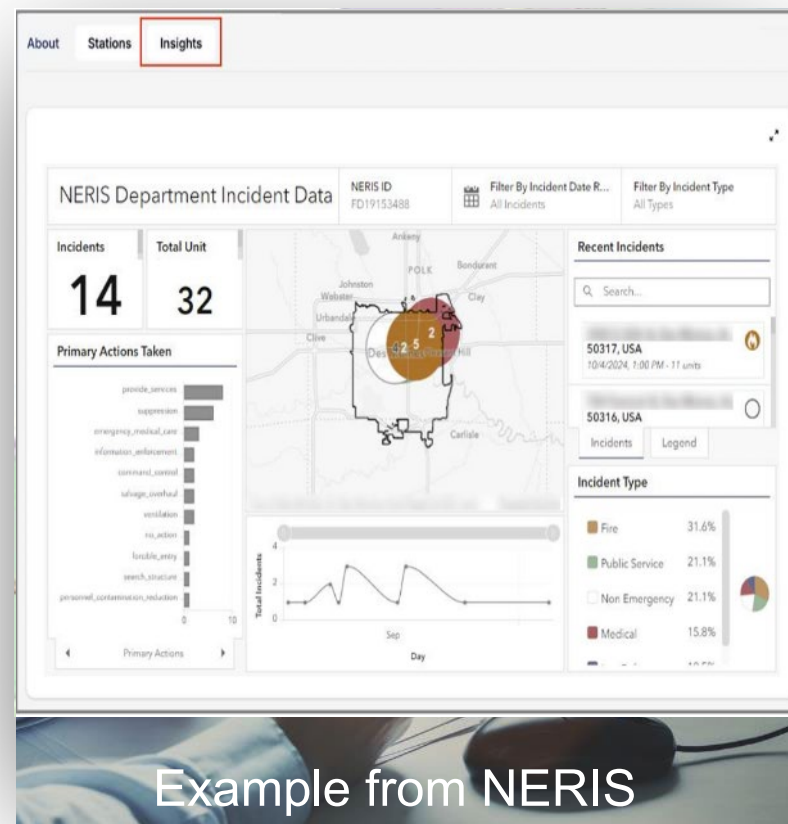
Report Clearly

Alright – we have the data. We’ve cleaned and analyzed it; now what?

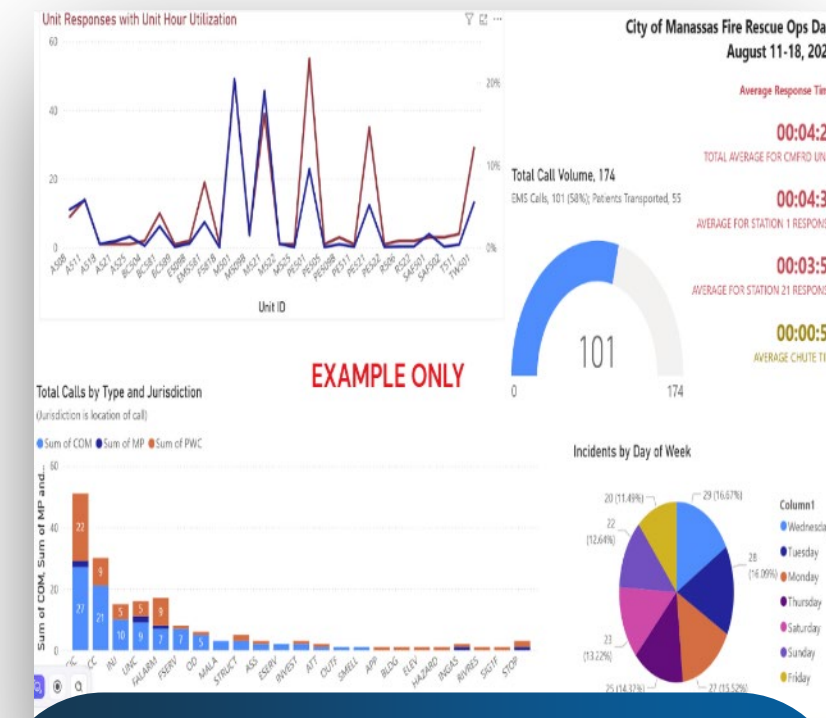
Now it’s time to report your findings. Ensure however you choose to report those findings that the information is clear and understandable so it can be used for insights and informed decision -making.



Example from First Due



Example from NERIS



And....an Example from Me
I'm still learning!

“What is a Call for Analyst’s Support”

	Life Cycle Element	Call for Service (CS)	Call for Analyst’s Support (CAS)
BEFORE THE CALL	Preparation	Equipment maintenance, education and training, operational readiness	Education and training, software
	Trigger	Emergency notification	Request for analysis or support
	Initial Response	Dispatch of units	Analyst engagement
DURING THE CALL	Tools	Apparatus, gear, SOPs	Software, study design/analytical frameworks, methods, and techniques
	On-Scene Adaptation	Tactical decisions	Change or adaptation of analytical methods and techniques, change of scope
AFTER THE CALL	Documentation	NFRIS, NERIS, AARs	Memos, reports, instructions, lessons identified
	Outcome	Incident resolution	Informed decision-making support

Key Takeaways

We've covered a lot of information today;
what should we take away from all this?



Set Clear Expectations

You and your supervisor need a clear understanding of what is being expected of you, and a timeline for achieving it.



Resources and Learning Opportunities

There are resources and learning opportunities everywhere. For example, if you aren't comfortable with large datasets yet, take a small amount of information you do know and follow the same steps – clean, analyze, and report it.



Be Patient, Have Grace with Yourself

Some people are built for this role, while others have to learn. Whichever you are, remember you are human and will make mistakes. It's OK. Get up and keep trying!

Question And Answer Session

Thank You For Your Attention



Melissa Heiderman

